Name: Daffa Gabillah Maulana Gapur
HVAC | Piping | Mechanical | Shipbuilding | Production Planner (LP Optimization)
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1. PORTFOLIO

1. GitHub: https://github.com/gabillah
2. VJudge: https://vjudge.net/user/daffaG
3. GrabCAD: https://grabcad.com/daffa.gabillah.maulana.gapur-1

2. LANGUAGE CERTIFICATION

1. (2026). TOEIC. Score: 665/990. Organizer: Politeknik Perkapalan Negeri Surabaya.
   Softfile certificate: https://github.com/gabillah/resume/blob/main/Language%20Certificate/TOEIC.pdf

3. EDUCATION BACKGROUND

1. Institution: Politeknik Perkapalan Negeri Surabaya.
   Study program: D4 Teknik Permesinan Kapal (Marine Mechanical Engineering).
   Period: August 2022 – July 2026.
   Location: Surabaya, Indonesia.
   Thesis: Analysis of Engine-Propeller Matching for a 160 TEUs Container Ship using B-Series and AU-Series Propellers.

4. WORK EXPERIENCE

1. Job position: Student Intern at PT PAL (Persero)
   Division: Design Division, Machinery Outfitting Department.
   Period: 21st February 2025 – 7th July 2025.
   Type of business: Shipbuilding.
   Location: Surabaya, Indonesia.
   Softfile certificate: https://github.com/gabillah/resume/blob/main/Work%20Experience%20Certificate/PT%20P
   Responsibilities:
   a. Designed and engineered ship piping systems in compliance with Lloyd's Register requirements, covering (2D & 3D) layout planning, firefighting, ballast, bilge, sewage treatment, fresh water and feed water cooling systems, fuel systems (MDO and HFO), compressed air, HVAC, refrigeration, and electrical systems. Achieved 15% reduction in total pipe length through optimized routing and sizing of all systems. Tools: manual calculations in spreadsheet; P&ID and technical drawings in AutoCAD; 3D machinery models in Rhinoceros; 3D piping in Revit; CFD validation in Pipe Flow Expert; Reporting using LaTex (TeXworks, MiKTeX, XeLaTeX + BibTeX).

5. PROJECT EXPERIENCE

1. (2026). Job-shop type manufacturing production planning. Linear programming optimization of operational/work-center/station/labor costs and work orders/lead time. Tools: Odoo; Python (scipy, pandas, openpyxl, matplotlib, & pulp) for optimization modeling; LaTeX (TeXworks, MiKTeX, XeLaTeX + BibTeX) for reporting.
2. (2025). Overhauled ship diesel engine. Tools: Performed measurements using a caliper, micrometer screw gauge, and dial bore gauge; Reporting using LaTex (TeXworks, MiKTeX, XeLaTeX + BibTeX).
3. (2025). Diesel fuel injection pressure testing. Adjust diesel fuel injection pressure according to product guide.
4. (2025). Conducted ship shaft alignment. Tools: Performed measurements using a dial gauge.
5. (2025). Designed an HVAC round ducting system with a 17,000 CFM capacity for the accommodation building of a 21,000 DWT oil tanker, in accordance with ASHRAE, CARRIER (static regain), and SNAME standards. Tools: Calculations performed using spreadsheet; Technical drawings produced using AutoCAD; Validated using Pipe Flow Expert software; Reporting using LaTex (TeXworks, MiKTeX, XeLaTeX + BibTeX).
   Portfolio:
   a. https://github.com/gabillah/ship-20k-dwt-137m-lpp-oil-product-tanker
6. (2025). Designed the propulsion and shaft system arrangement and performed Engine–Propeller Matching with a B-Series propeller for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawings using AutoCAD.
7. (2024). Performed steel welding operations using SMAW and aluminium welding operations using OAW.
8. (2024). Boiler Operator.
9. (2024). Carried out vacuum processing and R-22 refrigerant charging on HVAC systems.
10. (2023). Designed general arrangement and engine room arrangement drawings for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawing using AutoCAD.
    Portfolio:
    a. https://github.com/gabillah/ship-20k-dwt-137m-lpp-oil-product-tanker
11. (2023). Designed ship lines plan for a 21,000 DWT oil tanker. Tools: Calculations performed using spreadsheet; Technical drawing using AutoCAD.
    Portfolio:

   a. https://github.com/gabillah/ship-20k-dwt-137m-lpp-oil-product-tanker
12. (2022). Conducted non-destructive testing (NDT) (including magnetic particle testing, liquid penetrant testing, and ultrasonic testing) and destructive testing (DT) (including tensile testing, impact testing, and hardness testing) on welded alloy materials.
13. (2022). Lathe, milling, and drilling machine operator for pump shaft components.

6. VOLUNTEER EXPERIENCE

1. Job position: Staff Member at UKM Penalaran PPNS.
   Division: PKM Division.
   Period: September 2022 – December 2024.
   Type of organization: Educational organization.
   Location: Surabaya, Indonesia.
   Responsibilities:
   a. Served as Head of Events for Internal Monitoring and Evaluation, PKP2 Simulation, and PKP2:
      i. Drafted proposals, organized and executed events, and compiled activity reports.
      ii. Achieved 4 out of 5 teams advancing to the national finals. 1 team was unqualified as its members had already graduated prior to the official qualification.
   b. Served as Staff Member for the PIMPUS event:
      i. Reviewed 15 PKM proposals submitted by PPNS students.
      ii. Prepared event equipment.
   c. Served as Staff Member for the PIMNAS preparation team:
      i. Prepared event equipment.
2. Job position: Staff Member at Marine Engineering Student Association.
   Division: Communication and Information Division.
   Period: January 2023 – February 2024.
   Type of organization: Educational organization.
   Location: Surabaya, Indonesia.
   Responsibilities:
   a. Creating posts for the Marine Engineering department website
   b. Creating posters using Photoshop.
3. Job position: Graphic Design Volunteer at MILOO Project PT. Suryakata Teknologi Indonesia.
   Division: Design.
   Period: April 2021 – June 2021.
   Type of organization: Educational organization.
   Location: Surabaya, Indonesia.
   Responsibilities:
   a. Creating poster designs for talkshow events, using Figma, Photoshop, and Illustrator.
4. Job position: Coordinator of Graphic Design team (Volunteer) at Sense 2017–2018 (art festival organized by OSIS/MPK SMAN 6 Surabaya).
   Division: Design.
   Period: August 2017 – May 2018.
   Type of organization: Art festival.
   Location: Surabaya, Indonesia.
   Responsibilities:
   a. Creating Instagram feeds.
   b. Making posters.
   c. Documenting the event using a DSLR camera.
5. Job position: Staff of Graphic Design team (Volunteer) at Championsix 2016–2017 (sports festival organized by OSIS/MPK SMAN 6 Surabaya).
   Division: Design.
   Period: August 2016 – May 2017.
   Type of organization: Sport festival.
   Location: Surabaya, Indonesia.
   Responsibilities:
   a. Creating Instagram feeds.
   b. Making posters.
   c. Documenting the event using a DSLR camera.